SOURAV ROY

Mathematics & Algorithms → Quantitative Research & Data Science

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PROFILE

My background combines two separate TIFR master's degrees: most recently an M.Sc. in Computer and System Sciences (2021–2026) from TIFR STCS, Mumbai, preceded by an M.Sc. in Mathematics (2019–2021) from TIFR-CAM, Bengaluru. I study probability, randomized algorithms, convex geometry, computational mathematics and optimization, and develop reproducible Python/C++ experiments with explicit numerical limitations. I apply this mathematical and computational foundation to data science, financial time series, forecasting, portfolio analysis and transaction-risk modelling. My work emphasizes careful assumptions, chronological evaluation, leakage control, reproducibility and evidence-based interpretation. I am transitioning toward quantitative research, data science and risk analytics, including quantitative finance.

TECHNICAL SKILLS

Programming & data: Python, C++17, JavaScript / Node.js, NumPy, pandas, SciPy, scikit-learn, XGBoost, Matplotlib, CVXPY, Clarabel, SQL, SQLite; C, MATLAB, SageMath, Mathematica.

Quantitative & statistical methods: Probability, PCA, Ridge regression, Random Forest, KMeans, Gaussian HMMs, covariance estimation, minimum-variance optimization, risk forecasting, convex optimization, randomized algorithms and numerical simulation.

Research & evaluation: Chronological validation, precommitted holdout evaluation, fixed-cutoff forecasting, paired ranking and review-budget analysis, model calibration, feature ablation, backtesting with transaction costs, leakage auditing, forecast freezing, sampling, complexity analysis, numerical invariants, reproducibility checks, Python unittest and pytest, keyword-based RAG retrieval checks and source citation verification.

Tools: Git, GitHub, JupyterLab / Jupyter Notebook, VS Code, conda / Anaconda, macOS Terminal, DB Browser for SQLite, sqlite3 CLI, Homebrew, Apple Clang (C++17), LaTeX, LangChain.js and local Ollama/ Qwen3. These reflect tools I have used, with different levels of proficiency.

DATA SCIENCE, QUANTITATIVE FINANCE & RISK PROJECTS

Quant AI Market Lab | Independent quantitative research | completed v1.0.0 | 2026 | **Repository**

- Built a validated eight-ETF workflow covering returns, covariance, PCA, KMeans and Gaussian HMM market-condition models, Ridge and Random Forest forecasts of next-five-day SPY return RMS, and long-only minimum-variance portfolios.
- Verified return identities, covariance symmetry/PSD properties, eigendecomposition and explained variance. The first three PCs explained 79.01% of standardized historical return variation; on 565 common chronological dates Ridge MAE was 0.003201 versus 0.003443 for Random Forest.
- Compared five covariance methods, three transaction-cost assumptions, nine benchmark scenarios and 32 strategy-period robustness checks. Minimum-variance approaches reduced realized volatility in reported subperiods but underperformed SPY buy-and-hold on raw return; May-August 2026 are described as pseudo-out-of-sample rather than an untouched holdout.
- Compared 2-6 Gaussian HMM states, diagonal/full covariance options, multiple initializations, likelihood, AIC/BIC, occupancy and chronological diagnostics; retained an interpretable three-state full-covariance specification without claiming economically proven regimes. Examined Ridge coefficient stability across 87 monthly model refits.

Quant Market Lab II - Signal Processing & Prospective Forecasting | v0.1 baseline frozen; extensions in progress | 2026 | **Repository**

- Fixed the information cutoff at 31 August 2026; implemented Ridge, constant and rolling 20-day RMS baselines at four lead times, preserved 12 versioned September-December forecasts, and recorded model/data/source fingerprints with target-timing and prefix-invariance tests.
- Across 87 historical zero-gap windows Ridge had the lowest average absolute error in the repository comparison. September remains a retrospective check; nine October-December five-session forecasts were published from the frozen August-cutoff CSV without refitting. I implemented EWMA with separately frozen August/October information sets, then reproduced signed, squared and absolute-return autocorrelation without changing the forecasts. November and December outcomes remain pending; autoregressive extensions remain future work.
- Evaluated the first completed October five-session window: realized SPY daily-return RMS was 0.5263%, compared with the frozen rolling-20 forecast of 0.5780%, Ridge forecast of 0.8118% and constant baseline of 0.8791%. This is one evaluation window, not proof of predictive superiority.
- Reproduced August-data lag-one autocorrelations of -0.116 (signed), 0.420 (squared) and 0.352 (absolute returns). Verified the preserved original adjusted-price CSV against its SHA-256 manifest and documented differences between later download vintages. Kept reconstructed EWMA experiments separate from the original frozen forecasts.

Payment Guard | Completed synthetic fraud-review ranking study | 2026 | **Repository** · **Evidence and limits**

- Audited 1,090,000 synthetic AMLNet v2 transactions, reconciled 1,411 positive labels, inspected nested metadata and potential target leakage, prepared chronological splits, verified a SQLite import and investigated training-period patterns using SQL.
- Implemented regularized logistic regression and XGBoost with pandas/NumPy, scikit-learn preprocessing, chronological training and model selection, sigmoid/isotonic calibration, feature ablation and paired review-budget analysis. Frozen March–April 2026 final holdout: 253,190 transactions, 278 fraud labels; at the precommitted top-1% review capacity (2,532 reviews/model), XGBoost captured 267 positives versus logistic regression's 254 (+13 captured labels). Logistic captured 225 versus 222 at the smaller 0.1% budget; superiority was conditional on the chosen objective and workload.
- Analyzed feature sensitivity: XGBoost at 1% captured 169 positives without `category` and 141 without both `category` and `type` (267 with all features). Decision-time provenance of these synthetic columns remains unverified. Published aggregate-only plots and metrics, freeze and hash checks, and an auditable evaluation; **186 tests passed locally**. This is a completed synthetic-data research milestone, not a deployed bank fraud-detection system or established financial savings.

ResearchAudit | Evidence-based validation | rule-based audit plus tested local RAG/citation checks | 2026 | **Repository** · **Local LLM/RAG and citation tests**

- Froze Quant AI Market Lab v1.0.0 as an auditable source snapshot. The Python AST-based rule checker screened 34 files; 10 candidate locations were triaged and 52 test methods passed. Isolated Claim 7 transaction-cost tests supported specified cases while revealing two input-validation gaps; the complete portfolio backtest was not reproduced.
- Implemented and locally tested an experimental LangChain.js → Ollama → Qwen3 workflow with keyword-based RAG over three allowlisted source files. Retrieved passages carry file paths, line ranges and SHA-256 fingerprints; 13 JavaScript tests cover selected retrieval and citation checks. One recorded RAG answer recognized two already documented gaps, but manual review also found incomplete citations and an incorrect interpretation. This is not a semantic/vector search system, independent mathematical verification or a completed four-agent audit.

MATHEMATICAL & COMPUTATIONAL RESEARCH

Approximating Mixed Volumes to Arbitrary Accuracy | 2025 | Hariharan Narayanan and Sourav Roy | **arXiv paper**

- Co-authored a randomized approximation algorithm for mixed volumes of convex polytopes with integer vertices, combining convex optimization, Lorentzian polynomials and polytope subdivision; established multiplicative-error and failure-probability guarantees with polynomial runtime for a fixed number of distinct input polytopes under the stated assumptions.

Master's Thesis: Approximation of Mixed Volumes of Special Convex Bodies | TIFR Mumbai | March 2026 | Supervisor: Prof. Hariharan Narayanan | **Thesis PDF**

- Studied mixed volumes of a polytope and Euclidean ball using Minkowski sums, the Steiner polynomial, geometric decomposition, sampling and log-concavity; developed and analysed noisy binary search for a scaling parameter, including decision-error and runtime bounds, while distinguishing the idealized oracle model from practical approximate optimization.
- Implemented random-walk samplers for estimating volumes of convex bodies, including ball walk and hit-and-run variants, together with geometric membership/decomposition procedures and mixed-volume computations.

Exploring the Efficient Computation of Permanents of Graph Laplacians | TIFR | December 2022 | Supervisor: Prof. Piyush Srivastava | **Project report PDF**

- Proved the permanent analogue of Brenner's result for diagonally dominant matrices; explored tensor-product formulations, recursive expressions for paths, cycles and trees, and Pfaffian methods.

RSA Encryption and Decryption in Python | INSPIRE Summer Project, Presidency University Kolkata | 2019; revised 2026 | **Repository**

- Implemented educational RSA key generation and encryption/decryption using Miller-Rabin primality testing, the extended Euclidean algorithm and modular exponentiation. The 2026 revision adds UTF-8/ input validation and seven unit tests and documents correctness and public-key lookup attack limits of byte-wise textbook RSA.

Mixed Volume Approximations | Python and C++ numerical experiments | October 2026 | Repository

- I implemented numerical versions of the framework in my master's thesis, Approximation of Mixed Volumes of Special Convex Bodies (Chapters 4–6): B1 estimates volume, B2 samples a geometric component label, and B3 estimates component probabilities. Algorithms 1–3 organize sampling, scaling-parameter search and mixed-volume estimation. I checked the cube-plus-ball estimates against exact formulas and ran matched Python/C++ pilot comparisons. Finite burn-in, correlated samples and floating-point calculations do not establish the ideal-oracle guarantees or certify mixing and accuracy. A separate browser mixed-volume lab compares independent Monte Carlo estimates against exact cube references; it is not the thesis B1/B2/B3 oracle system.
- Compared matched Python/C++ pilot calculations across dimensions 7, 9 and 11 and scale parameters 0.5, 1.5 and 3.0; measured execution time and numerical error against exact cube cases, with reproducibility settings and limitations recorded in the repository.

Experiments on Random Walks | Computational convex geometry | Python/C++17 | October 2026 | **Repository · Interactive research website**

- Implemented Ball Walk, Dikin Walk, Coordinate Hit-and-Run, Random Hit-and-Run and a cube-specialized John's Walk on high-dimensional cubes, with readable explicit and optimized Python/C++17 code. Published source-labelled convergence checkpoints and cost comparisons for dimensions 1–10, with separate dimension-20 experiments. Distinguished empirical marginal/pairwise diagnostics from rigorous total-variation mixing; the browser experiment runner generates independent exploratory results while a separate archive searches curated past runs; browser timings are not native-language benchmarks.

EDUCATION

M.Sc., Computer and System Sciences | Tata Institute of Fundamental Research, School of Technology and Computer Science, Mumbai | 2021-2026 | First Class

M.Sc., Mathematics | Tata Institute of Fundamental Research, Centre for Applicable Mathematics, Bengaluru | 2019-2021 | First Class

B.Sc. (Honours), Mathematics | Serampore College, University of Calcutta | 2015-2018 | First Class

SELECTED COURSEWORK

Computer and System Sciences: Mathematical Foundations for Systems Sciences; Algorithms; Mathematical Logic; Cryptography; Advanced Probability; Polynomials and Applications; Topics in the Study of Markov Chains (half-course); Coding Theory; Topics in High Dimensional Geometry. Audited: Convex Optimization; Random Matrix Theory.

Mathematics: Linear Algebra; Algebra; Probability; Numerical Analysis; Real and Complex Analysis; Measure and Integration; Functional Analysis; Topology; Ordinary and Partial Differential Equations.

TIFR-CAM elementary PDE syllabus: method of characteristics; classification of second-order PDEs; Laplace, heat and wave equations; Green functions; maximum principles; separation of variables; and boundary-value solution methods. This is academic coursework, not a claim of professional derivatives-pricing experience.

PRESENTATIONS, ACADEMIC ACTIVITIES & RECOGNITION

- I spoke at the TIFR STCS volume-approximation seminar on 26 July 2024 and organized the Private Sequential Stateless Protocols seminar on 6 December 2024.
- Research presentation: Approximation of Mixed Volumes of Convex Bodies, STCS Symposium, TIFR, August 2025; seminar organization: Catalytic Graph Algorithms, STCS student seminar, TIFR, March 2026.
- Convex and Discrete Geometry Summer School, Alfréd Rényi Institute, Budapest, August-September 2023; School on Cluster Algebras, ICTS-TIFR, December 2018; Introduction to SageMath, ICT Mumbai and MTTS Trust, May-June 2020.
- TIFR / Department of Atomic Energy Research Scholarship, 2019-2026; DST INSPIRE Scholarship, 2015-2019.
- Qualified JAM Mathematics (2018) and Mathematical Statistics (2019); ISI M.Math admission test (2018); CMI M.Sc. Data Science written test (2019).
- Languages: Bengali, English and Hindi. References: three TIFR faculty referees available on request.